TASK 2 WRITING GUIDE

Write clearly, justify your choices, and keep every point linked to your own solution

This guide helps you write only what is needed for Task 2.

The examiner will already see:

- your code
- your database structure
- your testing evidence
- your versioning evidence

Your writing should explain the **important technical decisions** behind your solution.

Every sentence you write should answer these three questions:

- What did I choose?
- Why was this suitable?
- What does it improve for the user, the data, or the code?

If a sentence only describes something, improve it.

Use your own project throughout. Two students using this guide should still write differently.

Section 1 – Functional Prototype

What this section is for

This section explains how your solution works technically and why your main choices suit the scenario.

The examiner will already see evidence in:

- `index.html`
- `static/app.js`

- `docs/schema.sql`

You do not need to explain every feature.

Write **5 short sentences**.

Sentence 1 – Explain what JavaScript controls in your project

Ask yourself

- What does JavaScript control in my solution?
- Which part of the project would not work without it?
- Is it handling forms, events, validation, displayed data, or requests?

Then ask yourself

- Why is JavaScript suitable here?
- What does it improve?
- Does this show efficiency as well as function?

Useful words and technical terms to research

- interactive
- dynamic
- validation
- client-side
- event
- request
- responsive
- process
- display
- efficient
- logic

Sentence starters

- JavaScript is used to...
- In my solution, JavaScript controls...
- JavaScript helps the system by...

Final self-check

Have I explained what JavaScript does, why it is suitable, and what improves because of this choice?

Sentence 2 – Explain how Supabase stores your data

Ask yourself

- How is my data organised?
- Are any tables linked?
- Why is this structure useful?

Then ask yourself

- How does this help keep data accurate or organised?
- How does it reduce duplication or improve integrity?

Useful words and technical terms to research

- structured
- relational
- linked
- records
- database
- organised
- integrity
- constraint
- primary key
- foreign key
- unique
- check constraint

Sentence starters

- Supabase stores…
- PostgreSQL is used to…

- My data is organised using…

Final self-check

Have I explained why this database structure helps the solution work accurately and efficiently?

Sentence 3 – Explain how CRUD happens in your project

Ask yourself

- How are records added?
- How are records shown?
- How are records changed?
- How are records removed?

Then ask yourself

- Why does this help users keep data accurate?
- How does this keep the interface and stored data in sync?

Useful words and technical terms to research

- create
- retrieve
- update
- delete
- accurate
- current
- live
- synchronised
- API
- fetch
- request
- response

Sentence starters

- API requests allow…
- CRUD functions are used to…
- Records are updated by…

Final self-check

Have I linked CRUD to what users actually do in my system and what this improves?

Sentence 4 – Explain authentication

Ask yourself

- What does login protect?
- Which features need restricted access?
- What can only logged-in users do?

Then ask yourself

- Why is authentication important for this scenario?
- What risk does it reduce?

Useful words and technical terms to research

- authorised
- secure
- restricted
- protected
- access
- verified
- authentication
- authorisation
- account creation
- login

Sentence starters

- Authentication is used so...
- Users must log in before...
- Access is restricted because...

Final self-check

Have I explained what authentication protects and why that matters?

Sentence 5 – Explain one coding decision that improves quality

Choose one real example:

- validation
- trimming input
- required fields
- reusable function
- try/catch
- duplicate checks
- fallback handling

Ask yourself

- What stops incorrect data?
- What improves reliability?
- What improves consistency?
- What improves efficiency or maintainability?

Useful words and technical terms to research

- validated
- consistent
- reliable
- efficient
- robust
- prevented

- trim
- reusable
- error handling
- defensive programming

Sentence starters

- Input is checked before…
- I used validation so…
- Error handling improves…
- Reusable logic helps because…

Final self-check

Have I explained what improved because of this decision?

Section 2 – Code Organisation

What this section is for

This section explains how your code is organised so another developer could understand and maintain it.

The examiner will already see:

- file names
- comments
- functions
- folder structure

Write **4 short sentences**.

Sentence 1 – Explain naming choices

Ask yourself

- How have I named variables and functions?

- Is the naming consistent?
- Would another developer understand the purpose quickly?

Then ask yourself

- How does this help readability and maintenance?

Useful words and technical terms to research

- consistent
- clear
- readable
- meaningful
- recognisable
- camelCase
- naming convention

Final self-check

Have I explained how my naming helps another developer understand the code?

Sentence 2 – Explain file organisation

Ask yourself

- Why are files separated into folders?
- What belongs in each file?
- Why is `docs`, `static`, and the main page separated?

Then ask yourself

- How would this help another developer work on the project?

Useful words and technical terms to research

- organised
- structured

- separated
- maintainable
- modular
- directory
- file structure

Final self-check

Have I explained how my structure makes the project easier to navigate and maintain?

Sentence 3 – Explain reusable logic

Ask yourself

- Which functions are reused?
- Where have I avoided repetition?
- Have I kept repeated logic in one place?

Then ask yourself

- What does this improve?

Useful words and technical terms to research

- reusable
- efficient
- modular
- maintainable
- duplication
- function
- abstraction

Final self-check

Have I explained how reusable logic improves efficiency or maintainability?

Sentence 4 – Explain defensive programming

Ask yourself

- Where do I catch errors?
- What happens if something fails?
- How do I stop the program from breaking?

Then ask yourself

- How does this improve reliability or robustness?

Useful words and technical terms to research

- robust
- controlled
- reliable
- prevented
- defensive programming
- exception handling
- try/catch

Final self-check

Have I explained how my code handles problems in a controlled way?

Section 3 – User Experience

What this section is for

This section explains why your interface is easy to use and helps users complete tasks correctly.

The mark scheme rewards effective input handling, guidance, error messages, outputs, and robust handling of common and unexpected errors.

Write **4 short sentences**.

Sentence 1 – Explain navigation

Ask yourself

- Can users move through the system easily?
- Is the layout consistent?
- Can users find the main features quickly?

Then ask yourself

- How does this help users complete tasks successfully?

Useful words and technical terms to research

- clear
- intuitive
- consistent
- logical
- findable
- navigation
- layout

Final self-check

Have I explained how users can move around the system and find what they need?

Sentence 2 – Explain labels and guidance

Ask yourself

- How do users know what to enter?
- Are labels, headings, and prompts clear?

Then ask yourself

- How does this reduce mistakes or confusion?

Useful words and technical terms to research

- guidance
- instruction
- labelled
- clear
- prompt
- placeholder
- user guidance

Final self-check

Have I explained how the interface guides the user to complete tasks correctly?

Sentence 3 – Explain validation messages

Ask yourself

- What happens when users make mistakes?
- Do messages clearly explain what to fix?

Then ask yourself

- How does this improve accuracy and usability?

Useful words and technical terms to research

- feedback
- validation
- message
- corrective
- error message
- input handling

Final self-check

Have I explained how my feedback helps users recover from mistakes?

Sentence 4 – Explain unexpected input handling

Ask yourself

- What happens if unusual data is entered?
- What happens if a request fails or no records are found?

Then ask yourself

- How does this improve robustness?

Useful words and technical terms to research

- robust
- reliable
- fallback
- unexpected error
- graceful handling
- output

Final self-check

Have I explained how the system still works sensibly when something unexpected happens?

Section 4 – Accessibility, Legal Requirements and Security

What this section is for

This section explains how your solution follows recognised standards, protects users, and meets legal expectations.

The mark scheme rewards consistent and effective application of standards and guidelines in relation to accessibility, compatibility, legal and ethical considerations, plus effective security controls for confidentiality, integrity and availability.

Write **5 short sentences**.

Sentence 1 – Explain accessibility choices

Ask yourself

- What helps users read and use the interface clearly?
- Have I improved text, colour, layout, spacing, or keyboard use?

Then ask yourself

- Which recognised accessibility guidance does this connect to?

Useful words and technical terms to research

- WCAG
- contrast ratio
- accessible
- keyboard navigation
- focus state
- readable
- screen reader
- perceivable
- operable

Sentence starters

- I improved accessibility by…
- My interface supports accessibility through…
- This follows accessibility guidance because…

Final self-check

Have I explained what helps users access the system more easily?

Sentence 2 – Explain accessibility support features

Ask yourself

- Are form controls labelled clearly?
- Can images or controls be understood without relying only on visuals?
- Does my HTML support assistive technology?

Then ask yourself

- Who benefits from this and why?

Useful words and technical terms to research

- ARIA
- alt text
- label
- semantic HTML
- screen reader
- form control
- assistive technology

Sentence starters

- Form controls include…
- I used semantic HTML so…
- ARIA helps because…

Final self-check

Would a user with assistive technology benefit from this choice?

Sentence 3 – Explain legal and ethical requirements

Ask yourself

- Does my project collect or display personal data?
- Have I explained data use clearly?
- Have I avoided collecting unnecessary data?

Then ask yourself

- Which law or principle is relevant to my system?

Useful words and legislation to research

- GDPR
- UK GDPR
- Data Protection Act 2018
- consent
- privacy notice
- personal data
- data minimisation
- lawful basis
- retention
- legal and ethical considerations

Sentence starters

- GDPR is relevant because…
- Personal data is handled by…
- Privacy information is included so…

Final self-check

Have I linked legal requirements to the actual data used in my project?

Sentence 4 – Explain security controls

Ask yourself

- What protects stored data?
- What stops unsafe input?
- How are accounts protected?
- How are changes restricted to the right user?

Then ask yourself

- Which part of confidentiality, integrity, or availability does this improve?

Useful words and technical terms to research

- validation
- sanitisation
- authentication
- authorisation
- prepared query
- secure access
- confidentiality
- integrity
- availability
- password policy
- hashing

Sentence starters

- Input is protected by...
- Authentication improves security because...
- Validation helps by...

Final self-check

Have I explained which security risk is reduced and what part of the system is protected?

Sentence 5 – Explain compatibility

Ask yourself

- Does the system work across different screen sizes?
- Is layout still usable when resized?
- Does it still function in different browsers or devices?

Then ask yourself

- How does this improve access for a wider range of users?

Useful words and technical terms to research

- responsive
- viewport
- media query
- compatible
- consistent layout
- cross-browser
- device testing

Sentence starters

- My layout remains usable because…
- Responsive design helps…
- Compatibility was improved by…

Final self-check

Would this still work clearly on another device or screen size?

Section 5 – Testing Log (`testing-log.md`)

What this section is for

This file shows what you tested, what happened, and what improved.

The mark scheme expects test data to include normal, erroneous, and extreme data, and expects testing to inform the iterative development process. It also expects evidence that when errors are found, they are corrected and regression testing is used to check no new errors were introduced.

Each entry should show:

- what was tested
- expected result
- actual result
- pass or fail
- improvement if needed
- retest if changed

Ask yourself before each test entry

- What feature am I testing?
- What input am I using?
- Is this normal, erroneous, boundary, wrong, or absent data?
- What should happen?
- What actually happened?
- If it failed, what changed?
- Did I retest afterwards?

Useful words and technical terms to research

- expected
- actual
- validation
- failure
- corrected
- retested

- boundary
- erroneous
- regression testing
- test data

Final self-check

Does this test entry clearly show what was tested, what happened, what changed, and what improved?

Section 6 – Versioning Log (`versioning.md`)

What this section is for

This file shows how your project improved over time.

The mark scheme rewards thorough recording of notable changes, supported rationales, and effective use of versioning across the iterative development process.

Each entry should explain:

- what changed
- why it changed
- what impact it had

Strong version entries often connect directly to a previous testing issue.

Ask yourself before each version entry

- What did I add or change?
- Why was this needed?
- Was it caused by testing, review, or improvement?
- What became better afterwards?
- Am I making clear whether this version should be kept as the basis for the next iteration?

Useful words and technical terms to research

- implemented
- refined
- improved
- resolved
- impact
- stability
- iteration
- rationale
- review
- decision

Final self-check

Does the impact line clearly explain why this version matters and what improved because of it?

Section 7 – Asset Log

What this section is for

List anything you did not create yourself.

Include:

- asset name
- source
- licence
- purpose

Only include assets actually used.

Ask yourself

- Did I create this myself?
- If not, can I prove where it came from?

- Am I allowed to use it?
- Why is it included in my project?

Useful words and technical terms to research

- licence
- copyright
- royalty-free
- attribution
- source
- purpose

Final self-check

Could someone else clearly trace where this asset came from and why I used it?

Final writing check before submission

Before writing any sentence, ask:

- Is this linked to my own project?
- Have I explained why it helps?
- Have I explained what it improves for the user, the data, or the code?
- Would an examiner see matching evidence in my files?

Strong writing always matches visible evidence.

Final reminder

Do not try to sound complicated.

The strongest responses are:

- specific
- technical
- accurate
- brief

- clearly linked to your own solution